Lab Exam: Git & GitHub Workflow

MSIS DevOps

Duration: 90 minutes (practical) **Followed by:** Viva voce

Name: _____ **Reg. No.:** _____

GitHub username: _____ **Date:** _____

Instructions

- Perform all operations in order. Each step builds on the previous one.
- Use the Git command line for all local operations.
- Keep your terminal open. You may be asked to show your command history, `git log`, and the pull requests on GitHub during the viva.
- Make one commit per step that asks for a commit, with a meaningful commit message.
- The practical is followed by a viva on the commands you used and why.

Tasks

Perform the following Git operations:

1. Create a local repository called `msis-devops` containing a `readme.md` file, and push it to a **private** GitHub repository with the same name.
2. Create a new branch called `project-setup`. Set up a sample Python project containing `demo.py` and an appropriate `.gitignore` file.
3. Commit the changes, push the branch to the remote repository, and create a pull request (PR).
4. Merge the PR into `main`.
5. Switch to `main` and pull the latest changes from the remote repository. Create a new branch called `math-ops`. In `demo.py`, add functions named `add` and `sub`, each accepting two integers and returning the result.
6. Commit the changes, push the branch, create a PR, and merge it into `main`.
7. Switch to `main` and pull the latest changes. Create a new branch called `multiply-op` and implement a `multiply` function in `demo.py`. Commit the changes, push the branch, and create a PR. **Do not merge this PR yet.**
8. Switch back to `main` and create a new branch called `divide-op`. Implement a `divide` function in `demo.py`. Commit the changes, push the branch, and create a PR.
9. Merge the PR created in step 7 into `main`.
10. Switch to `divide-op` and pull the latest changes from the remote `main` branch into it.
11. Resolve any merge conflicts, ensuring that both the `multiply` and `divide` functions are retained in `demo.py`.
12. Commit the conflict resolution and push the updated `divide-op` branch to the remote repository.
13. Merge the PR created in step 8 into `main`.
14. Switch to `main`, pull the latest changes, and display `demo.py` to confirm that it contains all four functions: `add`, `sub`, `multiply`, and `divide`.

Examiner use only: Practical _____ Viva _____ Total _____ Sign _____